

3. One Step of Gradient Descent in a Deep Neural Network

Forward Pass

Input:

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Layer 1

$$z^{(1)} = W^{(1)}x + b^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$a^{(1)} = \text{ReLU}(z^{(1)}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Layer 2

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$a^{(2)} = \text{ReLU}(z^{(2)}) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Output Layer

$$s = W^{(3)}a^{(2)} + b^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Part A: Softmax with Cross-Entropy

Softmax definition:

$$q_i = \frac{e^{s_i}}{\sum_{j=1}^2 e^{s_j}}$$

$$q_1 = \frac{e^0}{e^0 + e^2} = \frac{1}{1 + 7.389} = 0.119$$

$$q_2 = 0.881$$

Cross-entropy loss:

$$L_{CE} = -\log(q_y)$$

True class $y = 1$:

$$L_{CE} = -\log(0.119) = 2.13$$

Gradient w.r.t Logits

For softmax + cross entropy:

$$\frac{\partial L}{\partial s_i} = q_i - y_i$$

$$\nabla_s L = \begin{bmatrix} 0.119 - 1 \\ 0.881 - 0 \end{bmatrix} = \begin{bmatrix} -0.881 \\ 0.881 \end{bmatrix} = \delta^{(3)}$$

Backpropagation

Layer 3

$$\nabla W^{(3)} = \delta^{(3)} (a^{(2)})^T$$

$$\nabla b^{(3)} = \delta^{(3)}$$

$$\nabla W^{(3)} = \begin{bmatrix} 0 & -1.762 \\ 0 & 1.762 \end{bmatrix}$$

$$\nabla b^{(3)} = \begin{bmatrix} -0.881 \\ 0.881 \end{bmatrix}$$

Layer 2

$$\delta^{(2)} = (W^{(3)})^T \delta^{(3)} \odot \text{ReLU}'(z^{(2)})$$

$$\delta^{(2)} = \begin{bmatrix} 0 \\ 0.881 \end{bmatrix}$$

$$\nabla W^{(2)} = \delta^{(2)} (a^{(1)})^T = \begin{bmatrix} 0 & 0 \\ 0.881 & 0.881 \end{bmatrix}$$

$$\nabla b^{(2)} = \begin{bmatrix} 0 \\ 0.881 \end{bmatrix}$$

Layer 1

$$\delta^{(1)} = (W^{(2)})^T \delta^{(2)} \odot \text{ReLU}'(z^{(1)}) = \begin{bmatrix} 0.881 \\ 0.881 \end{bmatrix}$$

$$\nabla W^{(1)} = \delta^{(1)} x^T = \begin{bmatrix} 0.881 & 0.881 \\ 0.881 & 0.881 \end{bmatrix}$$

$$\nabla b^{(1)} = \begin{bmatrix} 0.881 \\ 0.881 \end{bmatrix}$$

Gradient Descent Update

Update rule:

$$\theta \leftarrow \theta - \eta \nabla \theta$$

Learning rate $\eta = 0.1$

$$W_{\text{new}}^{(3)} = \begin{bmatrix} 1 & 0.2 \\ 0 & 0.8 \end{bmatrix}$$

$$b_{\text{new}}^{(3)} = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

$$W_{\text{new}}^{(2)} = \begin{bmatrix} 1 & -1 \\ 0.9 & 0.9 \end{bmatrix}$$

$$b_{\text{new}}^{(2)} = \begin{bmatrix} 0 \\ -0.1 \end{bmatrix}$$

$$W_{\text{new}}^{(1)} = \begin{bmatrix} 0.9 & -0.1 \\ -0.1 & 0.9 \end{bmatrix}$$

$$b_{\text{new}}^{(1)} = \begin{bmatrix} 0 \\ -0.1 \end{bmatrix}$$

Part B: Multiclass Hinge Loss

$$L_{\text{hinge}} = \sum_{j \neq y} \max(0, s_j - s_y + 1)$$

Since $y = 1$:

$$L_{\text{hinge}} = \max(0, s_2 - s_1 + 1) = \max(0, 2 - 0 + 1) = 3$$

Gradient w.r.t logits Because the margin is active,

$$\frac{\partial L}{\partial s_1} = -1, \quad \frac{\partial L}{\partial s_2} = 1$$

$$\delta^{(3)} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Layer 3

$$\nabla W^{(3)} = \delta^{(3)} (a^{(2)})^T = \begin{bmatrix} 0 & -2 \\ 0 & 2 \end{bmatrix}$$

$$\nabla b^{(3)} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Layer 2

$$\delta^{(2)} = (W^{(3)})^T \delta^{(3)} \odot \text{ReLU}'(z^{(2)})$$

Since $z^{(2)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$,

$$\delta^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\nabla W^{(2)} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\nabla b^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Layer 1

$$\delta^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\nabla W^{(1)} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\nabla b^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Gradient Descent Update ($\eta = 0.1$)

$$W_{\text{new}}^{(3)} = \begin{bmatrix} 1 & 0.2 \\ 0 & 0.8 \end{bmatrix}$$

$$b_{\text{new}}^{(3)} = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

$$W_{\text{new}}^{(2)} = \begin{bmatrix} 1 & -1 \\ 0.9 & 0.9 \end{bmatrix}$$

$$W_{\text{new}}^{(1)} = \begin{bmatrix} 0.9 & -0.1 \\ -0.1 & 0.9 \end{bmatrix}$$